(guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44

ANNUAL REVIEWS

Annual Review of Animal Biosciences

Ruminating on Bovine Implantation: Its Importance in Fertility, Food Production, Conservation, and Health

Olga Amelkina, Samantha Gardner, Jessica C. Edge, Zenab Butt, Haidee Tinning, and Niamh Forde

Discovery and Translational Sciences Department, Leeds Institute of Cardiovascular and Metabolic Medicine, School of Medicine, University of Leeds, Leeds, United Kingdom; email: n.forde@leeds.ac.uk

ANNUAL REVIEWS CONNECT

www.annualreviews.org

- Download figures
- Navigate cited references
- Keyword search
- Explore related articles
- Share via email or social media

Annu. Rev. Anim. Biosci. 2025. 13:143–65

First published as a Review in Advance on November 12, 2024

The *Annual Review of Animal Biosciences* is online at animal.annualreviews.org

https://doi.org/10.1146/annurev-animal-111523-102403

Copyright © 2025 by the author(s). This work is licensed under a Creative Commons Attribution 4.0 International License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited. See credit lines of images or other third-party material in this article for license information.

OPEN ACCESS

Keywords

endometrium, conceptus, receptivity, uterus, embryo, mammalian implantation

Abstract

Implantation in cattle is a key developmental checkpoint for pregnancy success. It involves careful spatiotemporal changes to the transcriptional landscape of the endometrium, with the heterogeneous nature of the endometrium increasing the complexity of understanding of the mechanism involved. Implantation is impacted by the developmental competency of the embryo, use of assisted reproductive technologies, and the environment in which this process occurs. We identify the factors that most impact the implantation process in cattle and highlight how it differs with that in other placental mammals. We propose the major areas that lack evidence are the mechanism(s) by which implantation itself occurs and how different stressors alter this process. Our understanding is hindered by a lack of appropriate in vitro models; however, development of novel 3D tools and available data sets will further elucidate the implantation process. Perhaps more importantly, this will develop methods to mitigate against these stressors to improve implantation success and offspring health.

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44

INTRODUCTION

Implantation is one of the key stages of pregnancy success in placental mammals, and in cattle, as in other species, it is the time point prior to which most pregnancy loss occurs (1). For successful implantation to occur, coordinated molecular interactions between the uterine endometrium and a developmentally competent conceptus (embryo and extraembryonic membranes) are required. The endometrium is a complex tissue composed mostly of stromal fibroblast-like cells and a monolayer of epithelial cells lining the uterine cavity. Endometrial epithelial cells can be categorized into luminal and glandular cells, with endometrial glands classified as either superficial or deep glands. Interspersed throughout the endometrium are different immune cells and blood cells, all of which are modified throughout the estrous cycle to facilitate the different stages of implantation, e.g., apposition, attachment, and (in some species) invasion (**Figure 1**).

The Role of Maternally Derived Steroid Hormones Estrogen and Progesterone

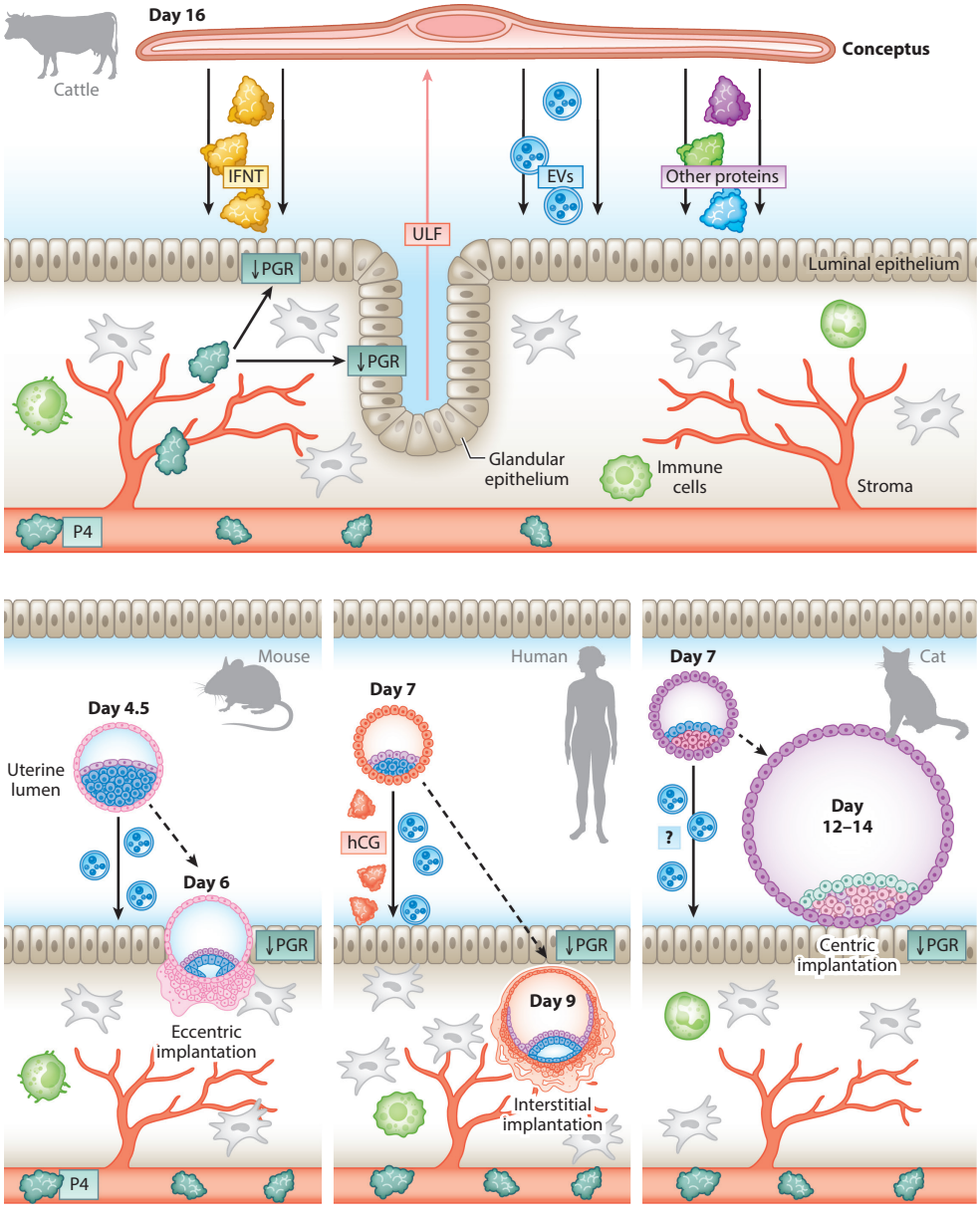
To understand the complexity of events required for successful implantation in cattle, we must first understand how the endometrium achieves a receptive state during the estrus cycle. Implantation in cattle occurs on days 21–23 post estrus, with the extraembryonic membranes adhering loosely to the endometrial epithelium (2). Establishment of receptivity is mediated primarily by the sequential actions of two maternal steroid hormones, estradiol (E2) and progesterone (P4), on the different endometrial cells throughout the cycle (3). In cattle, the estrus cycle is 18–24 days long, commencing at ovulation following release of an oocyte from the dominant follicle (3). Ovulation is triggered by follicle-derived E2 acting upon the hypothalamus, triggering an increased amplitude and frequency of luteinizing hormone surge from the anterior pituitary gland, initiating ovulation. E2 is secreted from the developing follicle and peaks approximately 36 h prior to ovulation in cattle (4). Mounting evidence indicates that E2 also has a role in endometrial receptivity in cattle; signs of estrus prior to artificial insemination are linked to both altered endometrial and conceptus gene expression, and increased conceptus length, compared to those displaying no signs of estrus prior to artificial insemination (5).

P4 exposure of the endometrium is a prerequisite to establishing endometrial receptivity to implantation in all mammalian species studied, including cattle (reviewed in 6, 7). Following ovulation, the follicle restructures into a corpus luteum (CL), which secretes increasing amounts of P4 until the CL degrades during a nonpregnant cycle (3). Increased P4 in maternal circulation advances the downregulation of the P4 receptor in luminal and superficial glandular endometrial cells (8). This loss of P4 receptor expression is considered the hallmark of a receptive endometrium, allowing a change in the transcriptional landscape of the endometrium to drive conceptus elongation, establish receptivity to implantation, and respond to the pregnancy recognition signal interferon tau (IFNT). Studies in cattle have found a correlation between maternal P4 concentration and conceptus length (9), which occurs indirectly via P4 actions upon the endometrium (10). Elongation sees the hatched blastocyst undergo a protracted period of rapid conceptus proliferation and elongation into a long filamentous conceptus and begins on days 12 to 14 in cattle (11). Longer conceptuses produce a greater amount of the pregnancy recognition signal IFNT (12), the consequences of which are discussed below.

In addition to its impact on the progesterone receptor, P4 also affects the global endometrial transcriptome. Transcriptome studies have identified that higher P4 concentrations alter the expression of genes involved in immune response, transport, metabolic processes, and embryo implantation on day 16 compared to day 13 (13). Recent work has explored the transcriptional response to P4 during estrus compared to day 7 post ovulation (14) and the specific response to P4 in different endometrial cell types (15). Increased P4 concentrations also affect uterine luminal

fluid (ULF) composition when compared to normal P4 concentration in cyclic heifers (16). ULF is the nutrition source for the developing conceptus, secreted primarily by the glandular epithelial cells into the uterine lumen with secretions known as histotroph. Altering ULF composition during early pregnancy by flushing the uterine lumen reduces implantation and pregnancy success (17). The histotrophic component of ULF is therefore a maternally derived source that contributes to conceptus development and successful implantation. Other sources, such as conceptus-derived factors, can also contribute to or alter implantation and pregnancy success.

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44



(Caption appears on following page)

Figure 1 (*Figure appears on preceding page*)

Overview of comparative aspects of implantation. Comparison of the peri-implantation period in cattle, mouse, human, and cat. Maternal steroid hormones, including P4, act upon the endometrium and contribute to advancing endometrial receptivity to implantation. One hallmark of endometrial receptivity is loss of the PGR in the luminal and superficial glandular epithelial cells. In cattle, the preimplantation blastocyst hatches, and the trophectoderm rapidly proliferates into a filamentous conceptus. By day 16, a competent conceptus secretes sufficient IFNT to trigger the MRP and modulate endometrial receptivity to implantation. The bovine conceptus also secretes other proteins and EVs that act upon the endometrium. This has also been shown in human and mouse. The bovine conceptus then attaches loosely to the luminal epithelial cells (centric implantation) by day 21. Human day 7 blastocysts secrete hCG to trigger MRP. No specific embryo-derived MRP signal is required in mouse or cat prior to implantation. The mechanism of implantation also differs across species. In mice, the blastocyst hatches and begins implantation, resulting in the embryo invading partly into the endometrium. The human hatched blastocyst follows the same process, but invasion into the endometrium is complete and involves the embryo burrowing completely into the stromal endometrial compartment. Finally, in cat, the blastocyst hatches and proliferates to fill the uterine lumen prior to superficial centric implantation on day 12–14. Abbreviations: EV, extracellular vesicle; hCG, human chorionic gonadotrophic hormone; IFNT, interferon tau; MRP, maternal recognition of pregnancy; P4, progesterone; PGR, progesterone receptor; ULF, uterine luminal fluid.

Embryo-Derived Factors that Enhance Receptivity

Successful implantation requires not only priming of the endometrial landscape by steroid hormones but also the coordinate development of a competent embryo/conceptus capable of signaling its presence to the mother. In cattle and other ruminants, IFNT (the pregnancy recognition signal) (18) is a type I interferon secreted in increasing concentrations from the conceptus trophectoderm during elongation (12). Validated in sheep, IFNT acts upon the luminal epithelium and superficial glands to inhibit oxytocin receptor expression, preventing prostaglandin synthase 2 release and thereby blocking luteolysis of the CL (19, 20). Maintenance of the CL results in extended P4 secretion, allowing the pregnancy to continue. Sufficient IFNT secretion by day 16 in cattle drives maternal recognition of pregnancy and precludes the process of implantation, which begins on days 19–20. The trophectoderm of the elongated conceptus aligns with the luminal epithelium and forms a loose attachment, mediated by adhesion molecules such as integrins (21). Simultaneously, the caruncular regions of the endometrium fuse with the conceptus cotyledons to form placentomes (22). IFNT's role in early pregnancy has been well characterized and reviewed extensively (23, 24). Its importance in implantation is via its distinct actions on different cellular components of the endometrium. It induces the expression of classical interferon-stimulated genes in the endometrium, such as ISG15, via binding to interferon alpha receptor 1/2, activation, and consequential downstream JAK/STAT signaling (25). ISG15 expression is present in (both superficial and deep glandular) luminal and stromal cells by day 18–23 (26). Although the role of many classical ISGs in early pregnancy is still under investigation, some, such as CCL8 and CXCL10, are thought to be involved in conceptus elongation, adhesion, or interaction with the endometrium (27, 28). Some interferon-stimulated genes are thought to be involved in modifying the immune response to the conceptus (28). IFNT also stimulates the expression of nonclassical ISGs (i.e., not stimulated by other type 1 interferons) that support conceptus elongation or implantation (reviewed in 29). Although less is known about nonclassical ISGs in cattle, studies in sheep have demonstrated that some nonclassical ISGs support elongation, including LGALS15, SLC2A1, and IGFBP1 (29). Other nonclassical ISGs, such as SPP1, are thought to be involved in conceptus attachment (30), whereas others are thought to modify the ULF to better support conceptus elongation (31). The endometrial transcriptional response to a conceptus is greater than to interferon exposure alone (32), indicating that the conceptus secretes other factors that influence the endometrium directly and may contribute to endometrial receptivity/implantation. The conceptus

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44

produces other factors such as proteins (33, 34), which can influence both the endometrial transcriptome (35) and extracellular vesicles in sheep (36) and cattle (37). Extracellular vesicles carry RNA species such as microRNAs and long noncoding RNAs, which can directly influence their target tissues (37, 38). Our understanding of the contribution of bovine conceptus-derived factors other than IFNT is limited, and relies on inferring information from other species such as sheep. To complicate this complex dialogue that occurs between conceptus and endometrium during this critical window of pregnancy recognition and implantation, other factors can also influence early pregnancy processes, such as paternally derived factors, the source of the embryo, and the environment in which the gametes and endometrium are exposed. Details of how these alter implantation and how we can develop tools to test the mechanism of implantation are expanded upon below.

Paternally derived influence on implantation. In addition to the main signaling mechanisms described above that are conserved (e.g., P4) or species/ungulate specific (e.g., IFNT), an additional component that can influence implantation is the male factor. The male ejaculate contains sperm cells and seminal plasma, which are deposited into different parts of the female reproductive tract depending on the species, e.g., the vagina in humans and cows or the uterus in mice and pigs (39). In vivo and/or in vitro responses of the female reproductive tract to seminal fluid have been shown in various species, including humans, mice, pigs, rabbits, golden hamsters, cows, sheep, horses and donkeys, dogs, and camelids (reviewed in 40, 41). Throughout development and ejaculation, sperm cells are exposed to the epididymis and male accessory sex glands. Consequentially, plasma contains secretions from both, and can alter the uterine environment and embryo development in a range of species (reviewed in 40, 42).

A specific focus is on how the male factor can influence the maternal immune response, with seminal fluid effects having been shown in vivo in cow (43), human (44), mouse (45), horse (46), pig (47, 48), and sheep (49), as well as in various in vitro cervical and endometrial cell systems, including cow (50, 51), human (52), mouse (45), and pig (53). In almost all species studied (except for golden hamster and camelids), seminal fluid induced neutrophil infiltration in the endometrium (40), and in cow (54, 55), mouse (56), pig (47), and golden hamster (57), seminal plasma affected embryo development.

Sperm small RNAs affect embryonic development and offspring phenotype in mice (58). In cattle and humans, where semen is deposited in the vagina, sperm migrate through the cervix into the uterus; this leaves most seminal plasma behind. In humans, semen mainly affects cervical lining (39). In horses and other mammals, both sperm and seminal plasma are deposited directly in the uterus (59). Interestingly, bovine sperm by itself showed low binding to neutrophils, but the addition of 10% seminal plasma increased sperm–neutrophil binding and the formation of neutrophil extracellular traps (59). In cattle, evidence suggests that the endometrial transcriptome differs when exposed to sperm from high- or low-fertility bulls (60) but not when exposed to seminal plasma from only vasectomized bulls (61), indicating the sperm may transport factors from the seminal plasma into the uterus. Conversely, exposure to seminal plasma 7 days prior to day 7 blastocyst transfer increased bovine conceptus length on day 14 (62). A study on intracervical deposition of seminal plasma in cows found no statistically significant effect of semen on the pregnancy rate (63). These data suggest that semen could alter the endometrium's ability to facilitate implantation either directly by altering the endometrium or indirectly via changes to conceptus length (**Figure 2**).

How unique is the implantation process in cattle? Our understanding of implantation mechanisms is enhanced when placed in an evolutionary context given the breadth of phenotypes involved in major pregnancy events in mammals. The morphological process of preimplantation is similar in most mammals, as the embryo develops into a blastocyst within the uterus.

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44

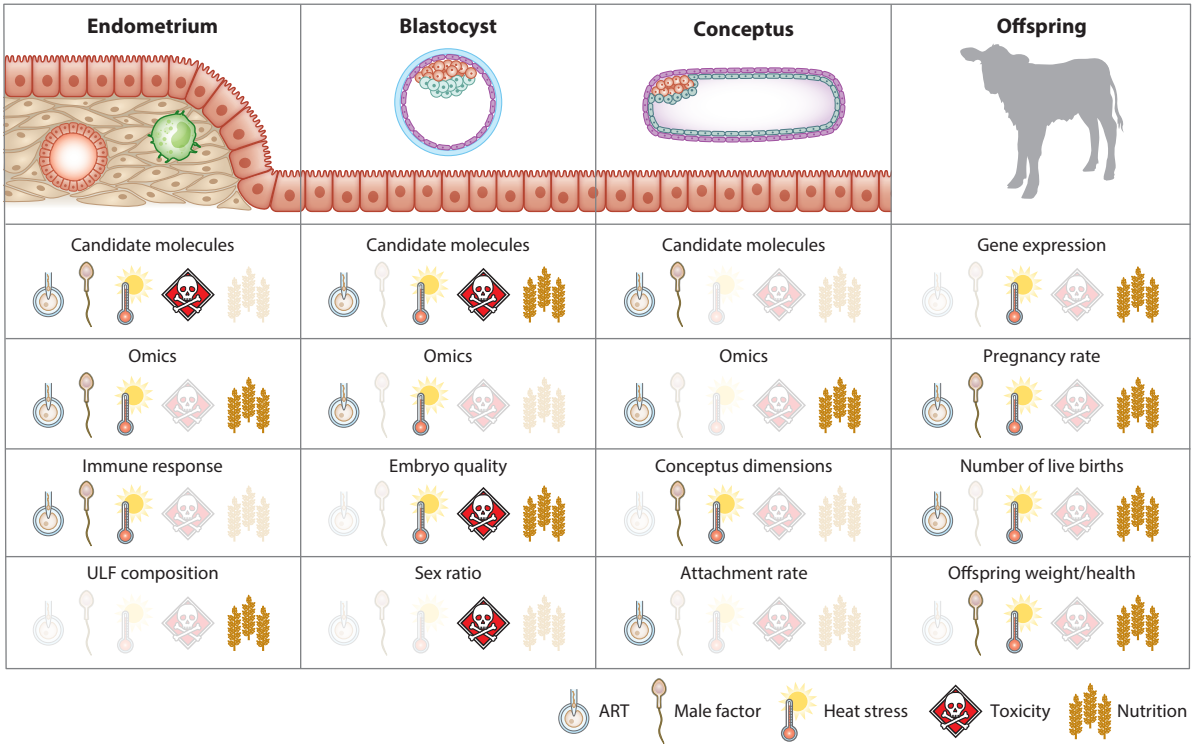


Figure 2

External factors that influence implantation, highlighting gaps in our knowledge where evidence is not available. Schematic diagram demonstrating the presence of evidence for the impact of ART, paternal influence, and environmental stressors (including heat stress and nutritional stressors) on different reproductive stages and outcomes. The influence of these factors on different physiological and molecular markers in the endometrium, blastocyst, peri-implantation conceptus, and offspring, and overall pregnancy success, is highlighted. A ghosted image indicates no evidence; gray image of sperm indicates a study was conducted but no effect was found. Abbreviations: ART, assisted reproductive technologies; ULF, uterine luminal fluid.

The type of implantation, however, can vary considerably, with domestic animals such as mares and ruminants displaying noninvasive implantation and primates, rodents, and carnivores showing an invasive type, in which the blastocyst migrates through the uterine luminal epithelia into stromal cells when decidualization takes place (64–66) (**Figure 1**). Within families, variations in implantation following adhesion and attachment also occur (67). Different types of implantation strategies have evolved in placental mammals; however, the ancestral implantation strategy from which these different implantation phenotypes arose from remains unresolved. Using a maximum likelihood model approach, one study concluded that superficial attachment implantation was the ancestral type for placental mammals, with interstitial attachment evolving in four mammalian orders (Rodentia, Primates, Chiroptera, and Eulipotyphla), a minimum of three times within rodents and at least twice in bats, Eulipotyphla, and insectivores (68).

From the embryo perspective, single-cell RNA sequencing of 6 mammalian species (human, cynomolgus monkey, marmoset, mouse, pig, and opossum), separated by up to 160 million years of evolution, has identified a switch in embryo metabolism between the blastocyst and gastrulation stages of development, which occurs independent of the type of implantation observed in these species (69). Although this study included no ruminants, a major difference in implantation morphology in cattle (and bovids) from the embryo side is in the formation of binucleate trophoblast

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44

cells found in all ruminant mammals. These cells migrate to fuse with the uterine epithelia, creating a feto-maternal hybrid tissue made up of individual syncytia, allowing a noninvasive type of implantation and placenta formation, and maintaining the maternal barrier (70).

Deer, sheep, and cattle form a feto-maternal syncytium at the preliminary implantation stages, with cattle specifically displaying tri-nucleated cells within this tissue. In cattle, it is rare but not unheard of for these tri-nucleated cells to fuse further with binucleated cells, forming multinucleated cells. From the syncytium, secretory granules containing intraluminal vesicles are released into the uterine epithelia. These vesicles serve the same purpose as exosomes, exocytosing trophoblast-derived proteins such as placental lactogen, prolactin-related protein, and pregnancy-associated glycoproteins, which are used for pregnancy diagnosis in these species. Detection of these proteins in the maternal endometrial stromal cells or through maternal circulation indicates deposit and functioning localized to these areas (71). Post granule release, these tri-nucleated cells die and are reabsorbed back into the trophectoderm (72). No other even-toed ungulates have been shown to have binucleated cells; Attiger et al. (73) suggest the development of these cells originated not from trophoblast cells but from stem cells located at the basal membrane. This is due to their consistent migration and fusion with the uterine epithelium, meaning they are constantly being replenished and renewed, with newly synthesized BNCs being localized within the basal membrane of the trophoblast (73). Collectively, these data indicate that bovid- and ungulate-specific components on both the maternal and the conceptus side facilitate this superficial type of implantation.

FACTORS THAT CAN INFLUENCE IMPLANTATION

How Does Embryo Source Alter Implantation in Cattle?

In addition to the required molecular changes that occur in the endometrium to facilitate implantation, the source of the embryo can have a substantial influence on implantation. Wiltbank et al. (74) showed recently that there are two waves of pregnancy loss in cattle, one in the preimplantation period and one in the peri-implantation period associated with implantation. These are associated with in vivo– and in vitro–produced embryos, whereas embryos derived from clones have pregnancy loss throughout the pregnancy; i.e., it is not limited to the pre- and peri-implantation period of pregnancy (75). Two seminal papers identified that embryos derived from in vivo or in vitro sources, i.e., those more likely to be developmentally competent, induce a substantially different transcriptional response in the endometrium compared to cloned embryos (those least developmentally competent) (21, 76). This biosensor component of the endometrium was also shown subsequently in other species such as humans. Further, this biosensor capability is misregulated in humans that experience recurrent implantation failure/early pregnancy loss (77). This transcriptional response has been demonstrated, along with the mechanism by which it may be mediated. Recent data have shown that it could be mediated via different microRNA cargo in extracellular vesicles derived from the conceptus itself during the peri-implantation period of pregnancy (37, 78, 79) (**Figure 2**). This demonstrates not only that the endometrium is required for successful implantation but also that its response to the developmental competence of the embryo plays a key role in implantation success.

Environmental Stressors and Their Impact on Implantation in Cattle

Paternal and maternal factors and the wider environment, including heat stress, nutritional status, and environmental pollutants, can influence the multifactorial process of implantation. In cattle, improved animal performance, measured by improved milk yields, animal growth, and reproduction, is influenced by environmental and animal husbandry factors, as extensively reviewed elsewhere (80). The effect of heat stress on oocytes in cows likewise has been reviewed extensively

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44

(81). In this section, we focus on the environmental stressors influencing the current state and future of bovine implantation research.

Paternal Stressors

As mentioned above, the male factor can affect implantation success and pregnancy outcomes in cattle; however, the major paternal environmental stressor that is a growing challenge is heat stress associated with climate change. Its consequences on implantation, pregnancy success, and offspring health are largely unknown but represent a major area requiring research (**Figure 2**). Ruminants testes are routinely measured at 2–6°C below body temperature, allowing for normal spermatogenesis. High seasonal temperatures are associated with detrimental changes in spermatogenesis (82), with heat stress causing decreased motility, a higher percentage of damaged sperm, and the decreased mitochondrial potential of ejaculated sperm (83). Contrary to this, bulls fed a high-nutrient diet have increased sperm production and therefore improved reproductive potential (84, 85). Once the sperm is deposited in the female reproductive tract, it induces an inflammatory response in the bovine endometrium, resulting in uterine clearance before implantation (86). The effect lower sperm quality would have on triggering these signaling pathways is an interesting question. In silico modeling of sperm migration, considering individual processes like rheotaxis and thermotaxis, could prove worthwhile for mechanism building (87). The heat shock proteins (HSPs) should be considered when assessing heat stress. HSPs may have a role in the early response to heat stress; plasma concentrations of HSP70 were demonstrated to be upregulated in steers in response to chronic temperature increases. Interestingly, these concentrations did not correlate with when the animal was exposed to the highest temperature, suggesting a decrease in any HSP-induced protective effects over time (88). Furthermore, in a porcine model, the HSP expression profile changes in response to heat stress (89). This demonstrates a pressing need to investigate the mechanisms through which paternal heat stress may influence implantation, pregnancy success, and offspring health and how these can be mitigated.

Maternal Stressors

From the maternal perspective, the lower probability of successful pregnancy and higher probability of embryonic loss associated with higher milk yield have been mostly rectified; however, dairy cows with high body temperatures have also shown pregnancy success and higher probability of embryonic loss (90). Interestingly, this increased body temperature is seen only in lactating cows and not in heifers, i.e., in animals in which early embryo loss is more frequent (91, 92). Offspring of heat-stressed dairy cows face consequences, with female offspring producing less milk per lactation and having lower body weight relative to dams provided with evaporative cooling (93). More recently, herd-level data have shown that offspring gestated in utero with elevated temperatures in the first or last three months of pregnancy are less robust across their life course (94). Indeed, heat stress effects can vary depending on the time of exposure to heat stress. Exposure during the first 16 gestational days results in lower conception rate and reduced conceptus weight (95). Furthermore, in utero exposure to heat stress over the last 6 weeks of gestation altered the DNA methylation profile of bull calf liver and heifer mammary gland (96). Transcriptome analysis on oocytes shows that heat stress can have direct effects in vitro, altering mitochondrial function and energy expenditure (97) and decreasing blastocyst production rates (98). A further global expression profile of bovine endometrium between the summer and autumn months showed significant differences in the transcriptional profile (99); specifically, heat stress alters the epidermal growth factor signaling pathway in the endometrium post estrus (100). The authors also noted decreased pregnancy rates after embryo transfer. Further in vitro studies demonstrated that heat stress in endometrial stromal cells increased production of prostaglandin F2α by tumor

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44

necrosis factor and altered chemokine production (101, 102). These studies implicate many potential players in the response to heat stress, but conclusive mechanisms remain elusive.

Fertility outcomes in the dairy cow are thought to be a balance between dietary input and energy expenditure (103). Negative energy balance has been associated with increased levels of nonesterified fatty acids, increased β-hydroxybutyrate, and low glucose concentrations. In fact, altered maternal metabolic status has a significant role in pregnancy success, with the transcriptome of the endometrium, blastocyst, conceptus, and ULF all involved (17, 97, 104–106). These studies suggest that the maternal environment provides crucial fetal programming for the subsequent offspring (107, 108). Using RNA-seq, Diniz et al. (84) recently studied vitamin and mineral supplementation strategies in heifers in the periconception and early gestation periods. Adamiak et al. (109) demonstrated altered liver size and implicated energy- and lipid-regulating pathways such as insulin/PI3K-Akt. In addition, increased body fat leads to a higher proportion of hyperinsulinemic animals, which is linked to impaired oocyte quality. These studies provide substantial evidence that maternal environment and reproductive outcomes are intrinsically linked, though the detailed mechanisms surrounding involvement in bovine implantation remain elusive.

Wider Environmental Stressors

A complication of increased global temperatures is the environmental exposure of cattle to these associated stressors, including heat stress. One method to compensate for lower reproductive outcomes in cattle is to incorporate cow breeds that are more resilient to heat stress or indeed, through gene introgression or gene editing, to modify expression of candidate genes that are associated with more resilience to heat stress (110, 111). For example, cattle breeds that carry the *SLICK1* allele of the prolactin receptor gene have improved thermotolerance, as demonstrated by lower body temperature (112). Higher global temperatures increase susceptibility to wildfires, leading to smoke inhalation by cattle. This increases the potential for exposure to toxins, such as dioxins, that may affect reproductive viability (113). Detailed studies on the effects of climate change on reproductive outcomes remain limited but are of increasing importance given the increased frequency and duration of extreme heat events.

The impact of human activity on reproductive outcomes cannot be underestimated. One example of this is the production of chemicals, such as polychlorinated biphenyls (PCBs) and polyfluoroalkyl substances, that resist degradation and so persist in the environment. PCBs decrease bovine oviductal contractions, and some showed increased leukocyte inhibitory factor mRNA synthesis during the estrous cycle (114). Furthermore, PCB exposure results in decreased prostaglandin secretion from the oviduct (115). Exposing bovine oocytes to perfluorohexane sulfonate induced phenotypic, transcriptomic, and DNA methylation changes in the resulting embryos (116). As exposure to many types of potential endocrine disruptors increases, in addition to the bioaccumulation effects, it will become pertinent to fully understand the reproductive outcomes. However, investigating bovine implantation is difficult due to the complexities that come with large animal science. As such, viable in vitro methods must be developed. Ultimately, the reproductive success of animals is intimately linked to human survival. Whether we are using animal models to better understand the human reproductive process or investigating reproduction with regard to food production, as we look to the future and the impending climate change disaster, we must understand how these temperatures will influence both animals and humans.

Using Bovine Implantation as a Proxy for Wild Species and Conservation Biology

Part of the Bovidae family of the order Cetartiodactyla, the domestic cow is an important model for assisting reproduction in wild ungulates. Wild ungulates are crucial to their ecosystems and

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44

are known as "nature architects," shaping habitats and interacting with the vegetation (117). The 345 identified wild ungulate species are classified mainly into the orders Cetartiodactyla and Perissodactyla. Most species from these orders are threatened or endangered due to habitat fragmentation or destruction, illegal hunting and trade, and disease (117–119). Captive populations of wild ungulates also face challenges because they are not self-sustaining (120). In this landscape, studying and applying reproductive sciences to wild ungulates is an essential part of the multidisciplinary work to conserve species (121–123). Enormous work has been done on developing and applying assisted reproductive techniques (ARTs) for wild species conservation, ranging from cloning (124–126), microfluidics (127), in vitro fertilization (IVF) (128), and embryo transfer (129) to contraception (130, 131) and cryopreservation (132, 133). However, despite substantial progress in understanding gamete production/storage and success of early embryo development, the environment these embryos require to develop in is not understood. Therefore, understanding of the major pathways and mechanisms of implantation in cattle, and which of these are conserved in other ungulates, can be used for conservation strategies for wild ungulates.

The bison is an example of successful application of ARTs in efforts to conserve a threatened member of the Bovidae family (134). North American bison (*Bison bison*) populations suffered a sharp decline from millions to a few hundred at the end of the nineteenth century due to unrestricted hunting. Through substantial progress in conservation efforts, this species' status has been changed from threatened to near threatened according to the International Union for Conservation of Nature's Red List (135). Substantial progress has been made in developing tools for in vitro production of bison embryos (136, 137), semen cryopreservation (138), and sequencing of the reference genome of the American plains bison (*Bison bison bison*) (139). The near threatened European bison (*Bison bonasus*) (140), the largest mammal in Europe, has benefitted from studies on its placenta and development of a germplasm bank (141, 142).

The banteng (*Bos javanicus*) is another member of the Bovidae family found in Southeast Asia and classified as endangered (143, 144). Promising studies have showed the possibility of interspecies nuclear transfer between banteng and cattle (145, 146). In addition, banteng reproductive activity can be monitored via noninvasive steroid hormone measurements (147), while mating systems may inform the biobanking strategies for this species (148). The gaur (*Bos gaurus*), listed as vulnerable (149), is another example of successful hybrid cloning with cattle (150–152).

Finally, an endangered member of the Bovidae family, the scimitar-horned oryx (*Oryx dammah*), was once widespread across North Africa but is now extinct in the wild, with surviving ex situ populations in Chad and the United States (153, 154). The reproductive and genetic management of these ex situ populations play a crucial role not only in saving the species from extinction but also in ongoing efforts to reintroduce the individuals back into their former range. The more we know about bovine implantation and pregnancy success, the more it will help inform conservation management and ARTs in endangered species.

How Can We Test the Mechanism of Implantation in Cattle?

We clearly have a good understanding of the main mechanism by which P4 and IFNT facilitate implantation in cattle, as well as of how different environment(s) or ARTs can influence the main morphological aspects of implantation, i.e., development of embryo/conceptus elongation or live pregnancy rates. However, information on the mechanisms by which different factors alter implantation in cattle is lacking. Two important approaches to address this inadequacy are modeling and generation of omics data, which can be used for various applications to unravel mechanisms (**Figure 3**). The main challenge to testing these mechanisms of action is the lack of tools to test hypotheses given the practical accessibility and ethical considerations for testing them in vivo. We therefore require appropriate in vitro models that mimic the uterine endometrium as well as

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44

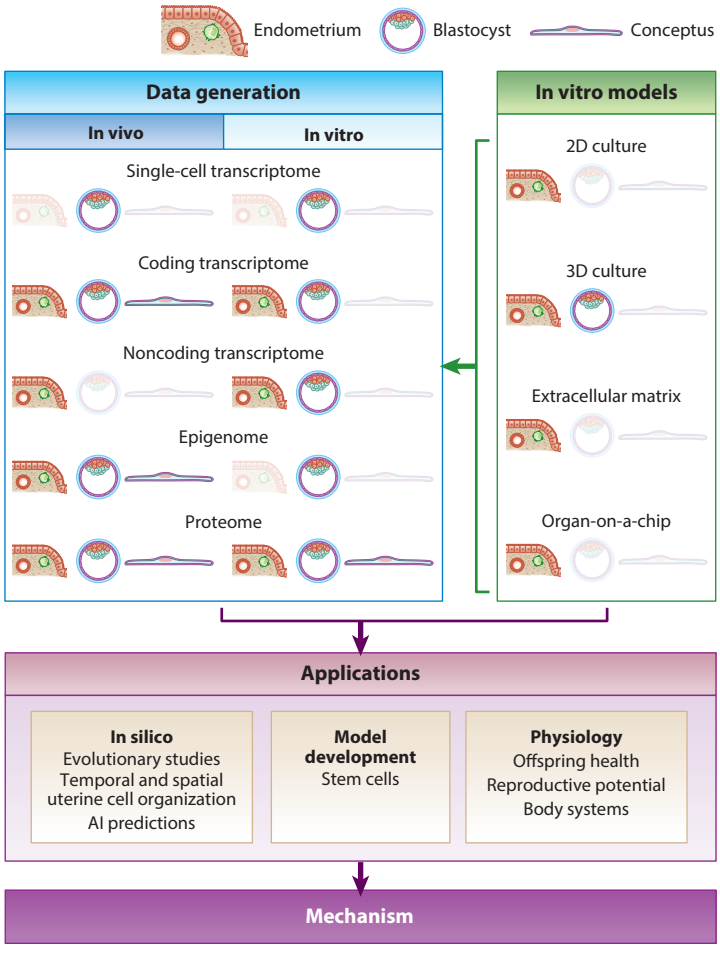


Figure 3

Data and tools are still required to understand the mechanism of implantation in cattle. In vitro models and in vivo–/in vitro–generated omics data that already exist for the bovine endometrium, blastocyst, and conceptus are depicted, and those still required for different applications to understand mechanisms behind bovine implantation are highlighted. A ghosted image indicates missing data and tools. Abbreviation: AI, artificial intelligence.

the embryo/conceptus to recapitulate these processes in vivo—similar to what has been demonstrated in other species (155). Recent advances in 3D embryo modeling have introduced blastoids as a tool for studying the bovine embryo (156). Previously achieved in mice (157) and humans (158), the generation of bovine blastoids provides a comparative species model for investigating the conservation or diversity of specific aspects of embryogenesis. Morphologically accurate embryo models were developed via use of expanded potential stem cells and trophoblast stem cells. These were confirmed following single-cell transcriptome analysis demonstrating that blastoids, at least at the transcriptomic level, are representative of in vivo embryos. These can survive in culture for two weeks and—following embryo transfer—produce the maternal recognition molecule IFNT. Prolonged study of bovine embryos in vitro is particularly relevant due to the post-hatching development previously discussed during the peri-implantation period, in which the conceptus

elongates prior to implantation. In vitro bovine blastocyst culture was limited mostly to days 8–9, the point at which elongation should begin, until efforts to extend this culture period and achieve elongation were carried out using agar gel tunnels to support growth (159–161). This method observed extended embryo survival and lengthening within tunnel walls, with evidence of an inner cell mass and differentiating cell groups; however, many experienced degeneration (160). Whether the observed elongation is due to true elongation or spatial restriction within the gel tunnels is unknown; however, results showed that the volume and area of nonelongated embryos were greater than those of the elongated ones within the tunnels, suggesting that tunnels limit growth (161). This system suggests that despite the requirement for gel tunnels to achieve structural lengthening, culture of spherical embryos could be carried out to day 15, and there was some evidence of processes important for elongation: trophoblast proliferation, hypoblast proliferation and migration, and epiblast development. However, pluripotency marker *SOX2* was absent from many of the embryos. This process is notoriously difficult to recreate in vitro, as endometrial secretions are thought to be required to drive elongation (30).

Historically, 2D, single cell–type in vitro static cultures have been used, which prove useful for some studies, for example, to treat bovine endometrial cells with important molecules such as CAPG to investigate functional effects (35). Unfortunately, these models lack many components of the in vivo endometrium and so are not entirely physiologically relevant. Furthermore, use of comparative species approaches, which provide some insight, is not always appropriate due to vast species variation in many aspects of early pregnancy, particularly implantation (155). Even culturing bovine uterine glands in 2D versus 3D culture using Matrigel altered cell morphology and hormone response between the culture techniques (162). The abundance of uterine gland secretions (as measured by mRNA) were significantly less expressed in 2D compared to 3D culture. This is likely due to differences in cell morphology, e.g., columnar-shaped cells with clear protein localization to apical and basolateral membranes in 3D culture compared to flat 2D culture with unclear spatial localization. Use of different 3D approaches is required, and recent advances in using scaffold-based structures have been successful. These Alvetex scaffolds—polystyrene inserts in which endometrial cells can be grown to form a tissue-like structure (163)—have been used previously for organs such as skin (164–166). In humans, this technology has been used to model the endocervix, which could be treated with menstrual cycle hormones over 28 days (167). Stromal and epithelial cells in this model responded appropriately to different cues at various stages of the menstrual cycle. This promising application could be used in multiple species to expand our understanding of species variation in early pregnancy. This method, applied to bovine endometrium, provided structural support to allow stromal cells to deposit extracellular matrix allowing the overlay of epithelial cells. This resulted in a physiologically relevant model (163) consisting of bovine endometrial stromal cells with an endometrial epithelial layer on top.

Microfluidic technology has been a leading method in the reproduction field, where endometrium-on-a-chip devices have been seeded with bovine stromal cells in a lower compartment and epithelial cells in a top compartment to alter glucose and insulin concentrations and observe the transcriptional effect on the endometrium (168). This technology has been extended further in other species, with development of a microfluidic device representative of the human menstrual cycle using mouse endometrial stromal and epithelial cells along with ovarian tissue, fallopian tube, ectocervix, and liver (169). This system showed prolonged P4 production following mimicked pregnancy signals. The benefit of this technology is that the individual chambers allow for movement of culture media and thus molecular crosstalk between the different cell types. This method could be a potentially promising option for coculture of endometrium and conceptus to investigate the peri-implantation period.

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44

Organoids are a recently developed technique to culture cells in 3D to represent a more complex structure. Turco et al. (170) generated hormone-responsive human endometrial glandular organoids that could be cultured in vitro long term. A negative aspect of this technology is the invasive nature of collecting these cells in humans; however, this can be mitigated by collection of tissue in animals destined for food sources or veterinary surgery waste. Furthermore, collection of tissue from menstrual blood, resulting in successful culture of organoids, has since been achieved (171), making this technology more accessible for menstruating species. This also allows for research into healthy donated endometrium, which would not be available normally. The more ready availability of endometrial tissue from bovine endometrium from the food production industry could create another avenue for species comparison. Currently, there is no evidence that bovine endometrial organoids have been cultured; however, organoids derived from other bovine tissues, for example, intestinal organoids, have been successful (172). Multicellular 3D spheroids to model the bovine endometrium have been produced, which improve the physiological relevance of the in vitro models (173). Additionally, bovine oviductal organoids have been established, along with those from dog, horse, cat, and pig (174), highlighting the exciting and promising prospects for this field in future early pregnancy research. Organoid research has also delved into modeling placental function, where it has been possible to culture human trophoblast organoids capable of differentiating into either extravillous trophoblasts or syncytiotrophoblasts (175). This work carried out in humans suggests that the technology is available to develop these methods in other species.

Single cell sequencing of bovine chorion was completed recently to investigate trophoblast and placental development at different stages of gestation (176). This elucidated the biological pathways and mechanisms controlling placental development, identifying candidate genes for further exploration. Single cell sequencing has also been conducted for human term placenta (177), allowing comparison of transcriptional activity between species to unravel the conservation and diversification of phenotypic placental characteristics. These studies emphasize the importance of generating omics data to assist in unraveling the mechanisms of implantation and early pregnancy, with single cell sequencing being a valuable tool. It has also been carried out on bovine blastocysts to investigate the effect of in vitro versus in vivo culture (178), demonstrating the differences introduced by culturing in vitro. Other omics data generation tools provide further insight that can be used to study different applications, such as physiological systems or in silico modeling (reviewed in 179). RNA sequencing to investigate the coding transcriptome has been performed on in vivo bovine endometrium (180), blastocyst (181), and conceptus (182), as well as in vitro endometrium models (35) and IVF blastocysts (183). However, the current lack of conceptus models in vitro makes this comparative approach impossible. RNA sequencing was also used on bovine endometrial biopsies to analyze the secretory proteins that may be present in ULF (184). Noncoding transcriptome investigation is less extensive, with microRNA expression studied in in vivo (185) and in vitro (186) bovine endometrium but not in the conceptus and only in IVF blastocysts (187). Epigenome data have been produced and explored in the endometrium (188) and conceptus (189) in vivo and comparatively both in vivo and in vitro in the blastocyst (190). To study translational activity, the proteome has been studied in the endometrium both in vivo (191) and in vitro (192) and in the blastocyst comparatively in vivo and in vitro (193). Conceptus-derived proteins were compiled from in vivo–produced conceptuses that were cultured in vitro for 6 h (34). Although these data sets have been generated, an inadequacy evidently remains in the available data to fully understand mechanisms in early pregnancy events. By combining accurate models, along with generating full data sets via omics approaches, we can strive to depict the mechanisms required for bovine implantation.

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44

CONCLUSION

Implantation in cattle is a tightly coordinated set of molecular interactions between the embryo and uterine endometrium. It is a key developmental time point in terms of pregnancy success; however, we still do not know enough about the required interactions and how these are affected by external stressors. Elucidating these mechanisms in cattle is important in its own right given how fundamentally interesting this process is for understanding mammalian pregnancy. Moreover, it is one of the key time points when pregnancy success or loss occurs. Maximizing pregnancy success is essential for more sustainable production of meat, milk, and milk-derived products, which is important in itself. It will also elucidate differences in implantation strategies in various mammalian species (**Figure 1**) and may facilitate conservation strategies for other bovid species. Many environmental stressors can impact bovine implantation (**Figure 2**), most pressingly those associated with climate change. However, knowledge of the mechanisms by which these alter implantation and subsequent life-course health of offspring is lacking. This is a consequence of lack of availability of appropriate models and data (**Figure 3**) to test mechanisms and, more importantly, mitigation strategies to ensure sustainable and healthy pregnancy in cattle and other mammalian species.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

ACKNOWLEDGMENTS

Research in N.F.'s lab is supported by BBSRC grant BB/X007332/1, Wellcome Trust grant 227178/Z/23/Z, and IAEA CRP: D31030 contract number 26223.

LITERATURE CITED

1. Tinning H, Edge JC, DeBem THC, Deligianni F, Giovanardi G, et al. 2023. Review: endometrial function in pregnancy establishment in cattle. *Animal* 17(Suppl. 1):100751
2. Brooks K, Burns G, Spencer TE. 2014. Conceptus elongation in ruminants: roles of progesterone, prostaglandin, interferon tau and cortisol. *J. Anim. Sci. Biotechnol.* 5:53
3. Forde N, Beltman ME, Lonergan P, Diskin M, Roche JF, Crowe MA. 2011. Oestrous cycles in *Bos taurus* cattle. *Anim. Reprod. Sci.* 124(3):163–69
4. Chenault JR, Thatcher WW, Kalra PS, Abrams RM, Wilcox CJ. 1975. Transitory changes in plasma progestins, estradiol, and luteinizing hormone approaching ovulation in the bovine. *J. Dairy Sci.* 58(5):709–17
5. Davoodi S, Cooke RF, Fernandes ACC, Cappellozza BI, Vasconcelos JLM, Cerri RLA. 2016. Expression of estrus modifies the gene expression profile in reproductive tissues on Day 19 of gestation in beef cows. *Theriogenology* 85(4):645–55
6. Bazer FW, Spencer TE, Johnson GA, Burghardt RC. 2011. Uterine receptivity to implantation of blastocysts in mammals. *Front. Biosci.* 3(2):745–67
7. Binelli M, Silva FACC, Rocha CC, Martins T, Sponchiado M, et al. 2022. Endometrial receptivity in cattle: the mutual reprogramming paradigm. *Anim. Reprod.* 19(4):e20220097
8. Okumu LA, Forde N, Fahey AG, Fitzpatrick E, Roche JF, et al. 2010. The effect of elevated progesterone and pregnancy status on mRNA expression and localisation of progesterone and oestrogen receptors in the bovine uterus. *Reproduction* 140(1):143–53
9. Carter F, Forde N, Duffy P, Wade M, Fair T, et al. 2008. Effect of increasing progesterone concentration from Day 3 of pregnancy on subsequent embryo survival and development in beef heifers. *Reprod. Fertil. Dev.* 20(3):368–75

10. Clemente M, de La Fuente J, Fair T, Al Naib A, Gutierrez-Adan A, et al. 2009. Progesterone and conceptus elongation in cattle: A direct effect on the embryo or an indirect effect via the endometrium? *Reproduction* 138(3):507–17
11. Betteridge KJ, Eaglesome MD, Randall GC, Mitchell D. 1980. Collection, description and transfer of embryos from cattle 10–16 days after oestrus. *J. Reprod. Fertil.* 59(1):205–16
12. Robinson RS, Fray MD, Wathes DC, Lamming GE, Mann GE. 2006. In vivo expression of interferon tau mRNA by the embryonic trophoblast and uterine concentrations of interferon tau protein during early pregnancy in the cow. *Mol. Reprod. Dev.* 73(4):470–74
13. Forde N, Carter F, Fair T, Crowe MA, Evans ACO, et al. 2009. Progesterone-regulated changes in endometrial gene expression contribute to advanced conceptus development in cattle. *Biol. Reprod.* 81(4):784–94
14. Alfattah MA, Correia CN, Browne JA, McGettigan PA, Pluta K, et al. 2024. Transcriptomics analysis of the bovine endometrium during the perioestrus period. *PLOS ONE* 19(3):e0301005
15. Pereira G, Guo Y, Silva E, Bevilacqua C, Charpigny G, et al. 2022. Progesterone differentially affects the transcriptomic profiles of cow endometrial cell types. *BMC Genom*. 23(1):82
16. Simintiras CA, Sánchez JM, McDonald M, Lonergan P. 2019. The influence of progesterone on bovine uterine fluid energy, nucleotide, vitamin, cofactor, peptide, and xenobiotic composition during the conceptus elongation-initiation window. *Sci. Rep.* 9(1):7716
17. Martins T, Pugliesi G, Sponchiado M, Gonella-Diaza AM, Ojeda-Rojas OA, et al. 2018. Perturbations in the uterine luminal fluid composition are detrimental to pregnancy establishment in cattle. *J. Anim. Sci. Biotechnol.* 9:70
18. Bazer FW, Spencer TE, Ott TL. 1997. Interferon tau: a novel pregnancy recognition signal. *Am. J. Reprod. Immunol.* 37(6):412–20
19. Spencer TE, Becker WC, George P, Mirando MA, Ogle TF, Bazer FW. 1995. Ovine interferon-tau inhibits estrogen receptor up-regulation and estrogen-induced luteolysis in cyclic ewes. *Endocrinology* 136(11):4932–44
20. Spencer TE, Ing NH, Ott TL, Mayes JS, Becker WC, et al. 1995. Intrauterine injection of ovine interferon-tau alters oestrogen receptor and oxytocin receptor expression in the endometrium of cyclic ewes. *J. Mol. Endocrinol.* 15(2):203–20
21. Mansouri-Attia N, Aubert J, Reinaud P, Giraud-Delville C, Taghouti G, et al. 2009. Gene expression profiles of bovine caruncular and intercaruncular endometrium at implantation. *Physiol. Genom.* 39(1):14–27
22. Pedersen HS, Mazzoni G, Stroebech L, Kadarmideen H, Hyttel P, Callesen H. 2017. Basic and practical aspects of pregnancy establishment in cattle. *Anim. Reprod.* 14:581–88
23. Bazer FW, Thatcher WW. 2017. Chronicling the discovery of interferon tau. *Reproduction* 154(5):F11–20
24. Bazer FW, Ying W, Wang X, Dunlap KA, Zhou B, et al. 2015. The many faces of interferon tau. *Amino Acids* 47(3):449–60
25. Stark GR, Kerr IM, Williams BR, Silverman RH, Schreiber RD. 1998. How cells respond to interferons. *Annu. Rev. Biochem.* 67:227–64
26. Austin KJ, Carr AL, Pru JK, Hearne CE, George EL, et al. 2004. Localization of ISG15 and conjugated proteins in bovine endometrium using immunohistochemistry and electron microscopy. *Endocrinology* 145(2):967–75
27. Sakumoto R, Hayashi K-G, Fujii S, Kanahara H, Hosoe M, et al. 2017. Possible roles of CC- and CXC-chemokines in regulating bovine endometrial function during early pregnancy. *Int. J. Mol. Sci.* 18(4):742
28. Nagaoka K, Nojima H, Watanabe F, Chang K-T, Christenson RK, et al. 2003. Regulation of blastocyst migration, apposition, and initial adhesion by a chemokine, interferon γ-inducible protein 10 kDa (IP-10), during early gestation. *J. Biol. Chem.* 278(31):29048–56
29. Hansen TR, Sinedino LDP, Spencer TE. 2017. Paracrine and endocrine actions of interferon tau (IFNT). *Reproduction* 154(5):F45–59
30. Spencer TE, Forde N, Lonergan P. 2016. The role of progesterone and conceptus-derived factors in uterine biology during early pregnancy in ruminants. *J. Dairy Sci.* 99(7):5941–50

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44

31. Mathew DJ, Sánchez JM, Passaro C, Charpigny G, Behura SK, et al. 2019. Interferon tau-dependent and independent effects of the bovine conceptus on the endometrial transcriptome. *Biol. Reprod.* 100(2):365–80
32. Bauersachs S, Ulbrich SE, Reichenbach H-D, Reichenbach M, Büttner M, et al. 2012. Comparison of the effects of early pregnancy with human interferon, alpha 2 (IFNA2), on gene expression in bovine endometrium. *Biol. Reprod.* 86(2):46
33. Forde N, Bazer FW, Spencer TE, Lonergan P. 2015. 'Conceptualizing' the endometrium: identification of conceptus-derived proteins during early pregnancy in cattle. *Biol. Reprod.* 92(6):156
34. Malo Estepa I, Tinning H, Rosas Vasconcelos EJ, Fernandez-Fuertes B, Sánchez JM, et al. 2020. Protein synthesis by day 16 bovine conceptuses during the time of maternal recognition of pregnancy. *Int. J. Mol. Sci.* 21(8):2870
35. Tinning H, Taylor A, Wang D, Constantinides B, Sutton R, et al. 2020. The role of CAPG in molecular communication between the embryo and the uterine endometrium: Is its function conserved in species with different implantation strategies? *FASEB J.* 34(8):11015–29
36. Burns GW, Brooks KE, Spencer TE. 2016. Extracellular vesicles originate from the conceptus and uterus during early pregnancy in sheep. *Biol. Reprod.* 94(3):56
37. De Bem THC, Bridi A, Tinning H, Sampaio RV, Malo-Estepa I, et al. 2023. Biosensor capability of the endometrium is mediated in part, by altered miRNA cargo from conceptus-derived extracellular vesicles. *FASEB J.* 38(10):e23639
38. Imakawa K, Matsuno Y, Fujiwara H. 2022. New roles for EVs, miRNA and lncRNA in bovine embryo implantation. *Front. Vet. Sci.* 9:944370
39. Miller DJ. 2024. Sperm in the mammalian female reproductive tract: surfing through the tract to try to beat the odds. *Annu. Rev. Anim. Biosci.* 12:301–19
40. Schjenken JE, Robertson SA. 2020. The female response to seminal fluid. *Physiol. Rev.* 100(3):1077–117
41. Paiva L, Silva M, Carrasco R, Ratto V, Goicochea J, Ratto M. 2022. Seminal plasma nerve growth factor signaling on the reproductive physiology of female llamas. *Anim. Reprod.* 19(4):e20220116
42. Bromfield JJ. 2016. A role for seminal plasma in modulating pregnancy outcomes in domestic species. *Reproduction* 152(6):R223–32
43. Howe GR, Black DL. 1963. Spermatozoan transport and leucocytic responses in the reproductive tract of calves. *J. Reprod. Fertil.* 6:305–11
44. Shen Q, Wu X, Chen J, He C, Wang Z, et al. 2023. Immune regulation of seminal plasma on the endometrial microenvironment: physiological and pathological conditions. *Int. J. Mol. Sci.* 24(19):14639
45. Sharkey DJ, Glynn DJ, Schjenken JE, Tremellen KP, Robertson SA. 2018. Interferon-gamma inhibits seminal plasma induction of colony-stimulating factor 2 in mouse and human reproductive tract epithelial cells. *Biol. Reprod.* 99(3):514–26
46. Woodward EM, Christoffersen M, Campos J, Betancourt A, Horohov D, et al. 2013. Endometrial inflammatory markers of the early immune response in mares susceptible or resistant to persistent breeding-induced endometritis. *Reproduction* 145(3):289–96
47. O'Leary S, Jasper MJ, Warnes GM, Armstrong DT, Robertson SA. 2004. Seminal plasma regulates endometrial cytokine expression, leukocyte recruitment and embryo development in the pig. *Reproduction* 128(2):237–47
48. Svensson A, Jiwakanon J, Fossum C, Dalin A-M. 2021. Expression of IL-23 in gilt endometrium and oviduct after insemination with seminal plasma, spermatozoa or semen extender. *BMC Res. Notes* 14(1):221
49. Scott JL, Ketheesan N, Summers PM. 2009. Spermatozoa and seminal plasma induce a greater inflammatory response in the ovine uterus at oestrus than dioestrus. *Reprod. Fertil. Dev.* 21(7):817–26
50. Ibrahim LA, Rizo JA, Fontes PLP, Lamb GC, Bromfield JJ. 2019. Seminal plasma modulates expression of endometrial inflammatory meditators in the bovine. *Biol. Reprod.* 100(3):660–71
51. Nongbua T, Guo Y, Ntallaris T, Rubér M, Rodriguez-Martinez H, et al. 2020. Bull seminal plasma stimulates in vitro production of TGF-β, IL-6 and IL-8 from bovine endometrial epithelial cells, depending on dose and bull fertility. *J. Reprod. Immunol.* 142:103179

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44

52. Sharkey DJ, Macpherson AM, Tremellen KP, Mottershead DG, Gilchrist RB, Robertson SA. 2012. TGF-β mediates proinflammatory seminal fluid signaling in human cervical epithelial cells. *J. Immunol.* 189(2):1024–35
53. Bai R, Latifi Z, Kusama K, Nakamura K, Shimada M, Imakawa K. 2018. Induction of immune-related gene expression by seminal exosomes in the porcine endometrium. *Biochem. Biophys. Res. Commun.* 495(1):1094–101
54. Stewart JL, Mercadante VRG, Dias NW, Canisso IF, Yau P, et al. 2018. Nerve growth factor-beta, purified from bull seminal plasma, enhances corpus luteum formation and conceptus development in *Bos taurus* cows. *Theriogenology* 106:30–38
55. Ortiz WG, Rizo JA, Carvalheira LR, Ahmed BMS, Estrada-Cortes E, et al. 2019. Effects of intrauterine infusion of seminal plasma at artificial insemination on fertility of lactating Holstein cows. *J. Dairy Sci.* 102(7):6587–94
56. Bromfield JJ, Schjenken JE, Chin PY, Care AS, Jasper MJ, Robertson SA. 2014. Maternal tract factors contribute to paternal seminal fluid impact on metabolic phenotype in offspring. *PNAS* 111(6):2200–5
57. Wong CL, Chan OC, Lee KH, O WS, Chow PH. 2008. Absence of paternal accessory sex glands dysregulates preimplantation embryo cell cycle and causes early oviductal-uterine transit in the golden hamster in vivo. *Fertil. Steril.* 89(4):1021–24
58. Yang C, Zeng Q-X, Liu J-C, Yeung WS-B, Zhang JV, Duan Y-G. 2023. Role of small RNAs harbored by sperm in embryonic development and offspring phenotype. *Andrology* 11(4):770–82
59. Alghamdi AS, Lovaas BJ, Bird SL, Lamb GC, Rendahl AK, et al. 2009. Species-specific interaction of seminal plasma on sperm-neutrophil binding. *Anim. Reprod. Sci.* 114(4):331–44
60. Donnellan EM, Cormican P, Reid C, Duggan G, Stiavnicka M, et al. 2023. The transcriptomic response of bovine uterine tissue is altered in response to sperm from high- and low-fertility bulls. *Biol. Reprod.* 108(6):912–21
61. Recuero S, Sánchez JM, Mateo-Otero Y, Bagés-Arnal S, McDonald M, et al. 2020. Mating to intact, but not vasectomized, males elicits changes in the endometrial transcriptome: insights from the bovine model. *Front. Cell Dev. Biol.* 8:547
62. Mateo-Otero Y, Sánchez JM, Recuero S, Bagés-Arnal S, McDonald M, et al. 2020. Effect of exposure to seminal plasma through natural mating in cattle on conceptus length and gene expression. *Front. Cell Dev. Biol.* 8:341
63. Odhiambo JF, Poole DH, Hughes L, Dejarnette JM, Inskeep EK, Dailey RA. 2009. Pregnancy outcome in dairy and beef cattle after artificial insemination and treatment with seminal plasma or transforming growth factor beta-1. *Theriogenology* 72(4):566–71
64. Bazer FW, Spencer TE, Johnson GA, Burghardt RC, Wu G. 2009. Comparative aspects of implantation. *Reproduction* 138(2):195–209
65. Stenhouse C, Seo H, Wu G, Johnson GA, Bazer FW. 2022. Insights into the regulation of implantation and placentation in humans, rodents, sheep, and pigs. *Adv. Exp. Med. Biol.* 1354:25–48
66. Paulson EE, Comizzoli P. 2021. Endometrial receptivity and embryo implantation in carnivores-commonalities and differences with other mammalian species. *Biol. Reprod.* 104(4):771–83
67. Chavan AR, Griffith OW, Stadtmauer DJ, Maziarz J, Pavlicev M, et al. 2021. Evolution of embryo implantation was enabled by the origin of decidual stromal cells in eutherian mammals. *Mol. Biol. Evol.* 38(3):1060–74
68. McGowen MR, Erez O, Romero R, Wildman DE. 2014. The evolution of embryo implantation. *Int. J. Dev. Biol.* 58(2–4):155–61
69. Malkowska A, Penfold C, Bergmann S, Boroviak TE. 2022. A hexa-species transcriptome atlas of mammalian embryogenesis delineates metabolic regulation across three different implantation modes. *Nat. Commun.* 13:3407
70. Wooding FBP. 2022. The ruminant placental trophoblast binucleate cell: an evolutionary breakthrough. *Biol. Reprod.* 107(3):705–16
71. Klisch K, Schraner EM. 2020. Intraluminal vesicles of binucleate trophoblast cell granules are a possible source of placental exosomes in ruminants. *Placenta* 90:58–61
72. Carter AM. 2020. Evolution of placentation in cattle and antelopes. *Anim. Reprod.* 16(1):3–17

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44

73. Attiger J, Boos A, Klisch K. 2018. Morphological characterization of basally located uninucleate trophoblast cells as precursors of bovine binucleate trophoblast giant cells. *Cells Tissues Organs* 205(3):151–63
74. Wiltbank MC, Monteiro PLJ, Domingues RR, Andrade JPN, Mezera MA. 2023. Review: maintenance of the ruminant corpus luteum during pregnancy: interferon-tau and beyond. *Animal* 17(Suppl. 1):100827
75. Wiltbank MC, Baez GM, Garcia-Guerra A, Toledo MZ, Monteiro PLJ, et al. 2016. Pivotal periods for pregnancy loss during the first trimester of gestation in lactating dairy cows. *Theriogenology* 86(1):239–53
76. Bauersachs S, Ulbrich SE, Zakhartchenko V, Minten M, Reichenbach M, et al. 2009. The endometrium responds differently to cloned versus fertilized embryos. *PNAS* 106(14):5681–86
77. Macklon NS, Brosens JJ. 2014. The human endometrium as a sensor of embryo quality. *Biol. Reprod.* 91(4):98
78. Nakamura K, Kusama K, Ideta A, Imakawa K, Hori M. 2020. IFNT-independent effects of intrauterine extracellular vesicles (EVs) in cattle. *Reproduction* 159(5):503–11
79. Nakamura K, Kusama K, Hori M, Imakawa K. 2023. Global analyses and potential effects of extracellular vesicles on the establishment of conceptus implantation during the peri-implantation period. *J. Reprod. Dev.* 69(5):246–53
80. Collier RJ, Dahl GE, VanBaale MJ. 2006. Major advances associated with environmental effects on dairy cattle. *J. Dairy Sci.* 89(4):1244–53
81. Roth Z. 2017. Effect of heat stress on reproduction in dairy cows: insights into the cellular and molecular responses of the oocyte. *Annu. Rev. Anim. Biosci.* 5:151–70
82. Shahat AM, Rizzoto G, Kastelic JP. 2020. Amelioration of heat stress-induced damage to testes and sperm quality. *Theriogenology* 158:84–96
83. Hamilton TRDS, Mendes CM, de Castro LS, de Assis PM, Siqueira AFP, et al. 2016. Evaluation of lasting effects of heat stress on sperm profile and oxidative status of ram semen and epididymal sperm. *Oxid. Med. Cell Longev.* 2016:1687657
84. Diniz WJS, Ward AK, McCarthy KL, Kassetas CJ, Baumgaertner F, et al. 2023. Periconceptual maternal nutrition affects fetal liver programming of energy- and lipid-related genes. *Animals* 13(4):600
85. Thundathil JC, Dance AL, Kastelic JP. 2016. Fertility management of bulls to improve beef cattle productivity. *Theriogenology* 86(1):397–405
86. Mansouri A, Yousef MS, Kowsar R, Usui N, Akthar I, Miyamoto A. 2023. Sperm activate TLR2/TLR1 heterodimerization to induce a weak proinflammatory response in the bovine uterus. *Front. Immunol.* 14:1158090
87. Diemer J, Hahn J, Goldenbogen B, Müller K, Klipp E. 2021. Sperm migration in the genital tract—in silico experiments identify key factors for reproductive success. *PLOS Comput. Biol.* 17(7):e1009109
88. Gaughan JB, Bonner SL, Loxton I, Mader TL. 2013. Effects of chronic heat stress on plasma concentration of secreted heat shock protein 70 in growing feedlot cattle. *J. Anim. Sci.* 91(1):120–29
89. Adur MK, Seibert JT, Romoser MR, Bidne KL, Baumgard LH, et al. 2022. Porcine endometrial heat shock proteins are differentially influenced by pregnancy status, heat stress, and altrenogest supplementation during the peri-implantation period. *J. Anim. Sci.* 100(7):skac129
90. Vasconcelos JLM, Demétrio DGB, Santos RM, Chiari JR, Rodrigues CA, Sá Filho OG. 2006. Factors potentially affecting fertility of lactating dairy cow recipients. *Theriogenology* 65(1):192–200
91. Sartori R, Sartor-Bergfelt R, Mertens SA, Guenther JN, Parrish JJ, Wiltbank MC. 2002. Fertilization and early embryonic development in heifers and lactating cows in summer and lactating and dry cows in winter. *J. Dairy Sci.* 85(11):2803–12
92. Rizos D, Carter F, Besenfelder U, Havlicek V, Lonergan P. 2010. Contribution of the female reproductive tract to low fertility in postpartum lactating dairy cows. *J. Dairy Sci.* 93(3):1022–29
93. Monteiro APA, Tao S, Thompson IMT, Dahl GE. 2016. In utero heat stress decreases calf survival and performance through the first lactation. *J. Dairy Sci.* 99(10):8443–50
94. Lewis K, Carter LS, Bradley A, Dewhurst R, Forde N, et al. 2024. Quantification of the effect of in-utero events on lifetime resilience in dairy cows. *J. Dairy Sci.* 107(7):4616–33
95. Kasimanickam R, Kasimanickam V. 2021. Impact of heat stress on embryonic development during first 16 days of gestation in dairy cows. *Sci. Rep.* 11:14839

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44

96. Skibiel AL, Dado-Senn B, Fabris TF, Dahl GE, Laporta J. 2018. In utero exposure to thermal stress has long-term effects on mammary gland microstructure and function in dairy cattle. *PLOS ONE* 13(10):e0206046
97. Sales JNS, Iguma LT, Batista RITP, Quintão CCR, Gama MAS, et al. 2015. Effects of a high-energy diet on oocyte quality and in vitro embryo production in *Bos indicus* and *Bos taurus* cows. *J. Dairy Sci.* 98(5):3086–99
98. Ferreira RM, Ayres H, Chiaratti MR, Ferraz ML, Araújo AB, et al. 2011. The low fertility of repeat-breeder cows during summer heat stress is related to a low oocyte competence to develop into blastocysts. *J. Dairy Sci.* 94(5):2383–92
99. Sakumoto R, Hayashi K-G, Saito S, Kanahara H, Kizaki K, Iga K. 2015. Comparison of the global gene expression profiles in the bovine endometrium between summer and autumn. *J. Reprod. Dev.* 61(4):297–303
100. Kawano K, Yanagawa Y, Nagano M, Katagiri S. 2022. Effects of heat stress on the endometrial epidermal growth factor profile and fertility in dairy cows. *J. Reprod. Dev.* 68(2):144–51
101. Sakai S, Yagi M, Fujime N, Kuse M, Sakumoto R, et al. 2021. Heat stress influences the attenuation of prostaglandin synthesis by interferon tau in bovine endometrial cells. *Theriogenology* 165:52–58
102. Sakai S, Hatabu T, Yamamoto Y, Kimura K. 2020. Alteration of chemokine production in bovine endometrial epithelial and stromal cells under heat stress conditions. *Physiol Rep*. 8(22):e14640
103. Leroy JLMR, Van Soom A, Opsomer G, Bols PEJ. 2008. The consequences of metabolic changes in high-yielding dairy cows on oocyte and embryo quality. *Animal* 2(8):1120–27
104. Bauersachs S, Simintiras CA, Sturmey RG, Krebs S, Bick J, et al. 2017. Effect of metabolic status on conceptus-maternal interactions on day 19 in dairy cattle: II. Effects on the endometrial transcriptome. *Biol. Reprod.* 97(3):413–25
105. Forde N, Simintiras CA, Sturmey RG, Graf A, Wolf E, et al. 2017. Effect of lactation on conceptus-maternal interactions at the initiation of implantation in cattle: I. Effects on the conceptus transcriptome and amino acid composition of the uterine luminal fluid. *Biol. Reprod.* 97(6):798–809
106. Moraes JGN, Behura SK, Geary TW, Spencer TE. 2020. Analysis of the uterine lumen in fertility-classified heifers: I. Glucose, prostaglandins, and lipids. *Biol. Reprod.* 102(2):456–74
107. Caton JS, Crouse MS, Reynolds LP, Neville TL, Dahlen CR, et al. 2019. Maternal nutrition and programming of offspring energy requirements. *Transl. Anim. Sci.* 3(3):976–90
108. Martin JL, Vonnahme KA, Adams DC, Lardy GP, Funston RN. 2007. Effects of dam nutrition on growth and reproductive performance of heifer calves. *J. Anim. Sci.* 85(3):841–47
109. Adamiak SJ, Mackie K, Watt RG, Webb R, Sinclair KD. 2005. Impact of nutrition on oocyte quality: cumulative effects of body composition and diet leading to hyperinsulinemia in cattle. *Biol. Reprod.* 73(5):918–26
110. Sarlo Davila KM, Hamblen H, Hansen PJ, Dikmen S, Oltenacu PA, Mateescu RG. 2019. Genetic parameters for hair characteristics and core body temperature in a multibreed Brahman-Angus herd. *J. Anim. Sci.* 97(8):3246–52
111. Hansen PJ. 2020. Prospects for gene introgression or gene editing as a strategy for reduction of the impact of heat stress on production and reproduction in cattle. *Theriogenology* 154:190–202
112. Carmickle AT, Larson CC, Hernandez FS, Pereira JMV, Ferreira FC, et al. 2022. Physiological responses of Holstein calves and heifers carrying the SLICK1 allele to heat stress in California and Florida dairy farms. *J. Dairy Sci.* 105(11):9216–25
113. Macias Franco A, Elins Moreira da Silva A, Holton G, Brody T, Alves Fonseca M. 2024. Establishing the relationship between wildfire smoke and performance metrics on finished beef cattle in Western Rangelands. *Transl. Anim. Sci.* 8:txae022
114. Wrobel MH, Mlynarczuk J. 2020. The effect of polychlorinated biphenyls (PCBs) on bovine oviductal contractions and LIF synthesis during estrous cycle, in vitro studies. *Res. Vet. Sci.* 133:188–93
115. Wrobel MH, Mlynarczuk J, Kotwica J. 2010. Influence of polychlorinated biphenyls and their hydroxylated metabolites on prostaglandins secretion from epithelial cells of bovine oviduct, in vitro. *Toxicology* 270(2–3):85–91

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44

116. Hallberg I, Persson S, Olovsson M, Moberg M, Ranefall P, et al. 2022. Bovine oocyte exposure to perfluorohexane sulfonate (PFHxS) induces phenotypic, transcriptomic, and DNA methylation changes in resulting embryos in vitro. *Reprod. Toxicol.* 109:19–30
117. Ripple WJ, Newsome TM, Wolf C, Dirzo R, Everatt KT, et al. 2015. Collapse of the world's largest herbivores. *Sci. Adv.* 1(4):e1400103
118. Berger J, Buuveibaatar B, Mishra C. 2013. Globalization of the cashmere market and the decline of large mammals in central Asia. *Conserv. Biol.* 27(4):679–89
119. Di Marco M, Collen B, Rondinini C, Mace GM. 2015. Historical drivers of extinction risk: using past evidence to direct future monitoring. *Proc. Biol. Sci.* 282(1813):20150928
120. Lees CM, Wilcken J. 2009. Sustaining the Ark: the challenges faced by zoos in maintaining viable populations. *Int. Zoo Yearb.* 43(1):6–18
121. Pukazhenthi BS. 2016. Saving wild ungulate diversity through enhanced management and sperm cryopreservation. *Reprod. Fertil. Dev.* 28(8):1133–44
122. Holt WV, Comizzoli P. 2022. Opportunities and limitations for reproductive science in species conservation. *Annu. Rev. Anim. Biosci.* 10:491–511
123. Roth TL, Swanson WF. 2018. From petri dishes to politics—a multi-pronged approach is essential for saving endangered species. *Nat. Commun.* 9:2588
124. Cowl VB, Comizzoli P, Appeltant R, Bolton RL, Browne RK, et al. 2024. Cloning for the twenty-first century and its place in endangered species conservation. *Annu. Rev. Anim. Biosci.* 12:91–112
125. Swegen A, Appeltant R, Williams SA. 2023. Cloning in action: Can embryo splitting, induced pluripotency and somatic cell nuclear transfer contribute to endangered species conservation? *Biol. Rev. Camb. Philos. Soc.* 98(4):1225–49
126. Borges AA, Pereira AF. 2019. Potential role of intraspecific and interspecific cloning in the conservation of wild mammals. *Zygote* 27(3):111–17
127. Le Gac S, Ferraz M, Venzac B, Comizzoli P. 2021. Understanding and assisting reproduction in wildlife species using microfluidics. *Trends Biotechnol.* 39(6):584–97
128. Paulson RJ, Comizzoli P. 2018. Addressing challenges in developing and implementing successful in vitro fertilization in endangered species: an opportunity for humanity to "give back." *Fertil. Steril.* 109(3):418–19
129. Comizzoli P. 2021. Adding new ingredients to the recipe for successful embryo transfers. *J. Assist. Reprod. Genet.* 38(5):1019–20
130. Patton ML, Jöchle W, Penfold LM. 2007. Review of contraception in ungulate species. *Zoo Biol.* 26(4):311–26
131. Sontakke SD. 2018. Monitoring and controlling ovarian activities in wild ungulates. *Theriogenology* 109:31–41
132. da Silva AM, Pereira AF, Comizzoli P, Silva AR. 2020. Cryopreservation and culture of testicular tissues: an essential tool for biodiversity preservation. *Biopreserv. Biobank.* 18(3):235–43
133. Campos LB, Praxedes ÉCG, Saraiva MVA, Comizzoli P, Silva AR. 2019. Advances and challenges of using ovarian preantral follicles to develop biobanks of wild mammals. *Biopreserv. Biobank.* 17(4):334–41
134. Acevedo C, Barfield JP. 2023. Review: reproductive physiology of bison and application of assisted reproductive technologies to their conservation. *Animal* 17(Suppl. 1):100842
135. Aune K, Jørgensen D, Gates C. 2016. *Bison bison*. In *IUCN Red List of Threatened Species*. Gland, Switz.: Int. Union Conserv. Nat.
136. Barfield JP. 2019. In vitro production of bison embryos. *Methods Mol. Biol.* 2006:165–77
137. Benham HM, McCollum MP, Nol P, Frey RK, Clarke PR, et al. 2021. Production of embryos and a live offspring using post mortem reproductive material from bison (*Bison bison bison*) originating in Yellowstone National Park, USA. *Theriogenology* 160:33–39
138. Hussain SA, Lessard C, Anzar M. 2011. Quantification of damage at different stages of cryopreservation of endangered North American bison (*Bison bison*) semen and the effects of extender and freeze rate on post-thaw sperm quality. *Anim. Reprod. Sci.* 129(3–4):171–79
139. Oppenheimer J, Rosen BD, Heaton MP, Vander Ley BL, Shafer WR, et al. 2021. A reference genome assembly of American bison, *Bison bison bison*. *J. Hered.* 112(2):174–83

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44

140. Plumb G, Kowalczyk R, Hernandez-Blanco JA. 2020. *Bison bonasus*. In *IUCN Red List of Threatened Species*. Gland, Switz.: Int. Union Conserv. Nat.
141. Kiewisz J, Melo de Sousa N, Beckers JF, Panasiewicz G, Gizejewski Z, Szafranska B. 2009. Identification of multiple pregnancy-associated glycoproteins (PAGs) purified from the European bison (Eb; *Bison bonasus* L.) placentas. *Anim. Reprod. Sci.* 112(3–4):229–50
142. Duszewska AM, Baraniewicz-Kołek M, Wojdan J, Barłowska K, Bielecki W, et al. 2022. Establishment of a wisent (*Bison bonasus*) germplasm bank. *Animals* 12(10):1239
143. Gardner P, Hedges S, Pudyatmoko S, Gray TNE, Timmins RJ. 2014. *Bos javanicus*. In *IUCN Red List of Threatened Species*. Gland, Switz.: Int. Union Conserv. Nat.
144. Purwantara B, Noor RR, Andersson G, Rodriguez-Martinez H. 2012. Banteng and Bali cattle in Indonesia: status and forecasts. *Reprod. Domest. Anim.* 47(Suppl. 1):2–6
145. Sansinena MJ, Hylan D, Hebert K, Denniston RS, Godke RA. 2005. Banteng (*Bos javanicus*) embryos and pregnancies produced by interspecies nuclear transfer. *Theriogenology* 63(4):1081–91
146. Steklenev EP, Elistratova TM. 1992. The characteristics of the reproductive capacity of hybrids of banteng (*Bos* (*Bibos*) *javanicus* d'Alton) with the domestic cow (*Bos* (*Bos*) *primigenius taurus*). *Tsitol. Genet.* 26(6):45–57, 75
147. Kozlowski CP, Bauman KL, Clawitter HL, Hall R, Poelker C, et al. 2022. Noninvasive monitoring of steroid hormone production and activity of zoo-housed banteng (*Bos javanicus*). *Anim. Reprod. Sci.* 247:107070
148. Gillis JD, Holt WV, Yon L, Woad KJ, Love D, et al. 2022. How can mating systems inform future biobanking strategies? An illustration using two Indonesian bovids, banteng (*Bos javanicus*) and lowland anoa (*Bubalus depressicornis*). *Anim. Reprod. Sci.* 238:106943
149. Duckworth JW, Sankar K, Williams AC, Samba Kumar N, Timmins RJ. 2016. *Bos gaurus*. In *IUCN Red List of Threatened Species*. Gland, Switz.: Int. Union Conserv. Nat.
150. Srirattana K, Imsoonthornruksa S, Laowtammathron C, Sangmalee A, Tunwattana W, et al. 2012. Full-term development of gaur-bovine interspecies somatic cell nuclear transfer embryos: effect of trichostatin A treatment. *Cell. Reprogr.* 14(3):248–57
151. Imsoonthornruksa S, Srirattana K, Phewsoi W, Tunwattana W, Parnpai R, Ketudat-Cairns M. 2012. Segregation of donor cell mitochondrial DNA in gaur-bovine interspecies somatic cell nuclear transfer embryos, fetuses and an offspring. *Mitochondrion* 12(5):506–13
152. Mastromonaco GF, Favetta LA, Smith LC, Filion F, King WA. 2007. The influence of nuclear content on developmental competence of gaur × cattle hybrid in vitro fertilized and somatic cell nuclear transfer embryos. *Biol. Reprod.* 76(3):514–23
153. Humble E, Stoffel MA, Dicks K, Ball AD, Gooley RM, et al. 2023. Conservation management strategy impacts inbreeding and mutation load in scimitar-horned oryx. *PNAS* 120(18):e2210756120
154. Woodfine T, Gilbert T. 2016. The fall and rise of the scimitar-horned oryx. In *Antelope Conservation*, ed. J Bro-Jørgensen, DP Mallon, pp. 280–96. Hoboken, NJ: John Wiley & Sons
155. Fitzgerald HC, Schust DJ, Spencer TE. 2021. In vitro models of the human endometrium: evolution and application for women's health. *Biol. Reprod.* 104(2):282–93
156. Pinzón-Arteaga CA, Wang Y, Wei Y, Ribeiro Orsi AE, Li L, et al. 2023. Bovine blastocyst-like structures derived from stem cell cultures. *Cell Stem Cell* 30(5):611–16.e7
157. Rivron NC, Frias-Aldeguer J, Vrij EJ, Boisset J-C, Korving J, et al. 2018. Blastocyst-like structures generated solely from stem cells. *Nature* 557(7703):106–11
158. Yu L, Wei Y, Duan J, Schmitz DA, Sakurai M, et al. 2021. Blastocyst-like structures generated from human pluripotent stem cells. *Nature* 591(7851):620–26
159. Brandão DO, Maddox-Hyttel P, Løvendahl P, Rumpf R, Stringfellow D, Callesen H. 2004. Post hatching development: a novel system for extended in vitro culture of bovine embryos. *Biol. Reprod.* 71(6):2048–55
160. Vajta G, Alexopoulos NI, Callesen H. 2004. Rapid growth and elongation of bovine blastocysts in vitro in a three-dimensional gel system. *Theriogenology* 62(7):1253–63
161. Ramos-Ibeas P, Lamas-Toranzo I, Martínez-Moro Á, de Frutos C, Quiroga AC, et al. 2020. Embryonic disc formation following post-hatching bovine embryo development *in vitro*. *Reproduction* 160(4):579–89

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44

162. Sugino Y, Sato T, Yamamoto Y, Kimura K. 2022. Evaluation of bovine uterine gland functions in 2D and 3D culture system. *J. Reprod. Dev.* 68(4):254–61
163. Díez MC, Przyborski S, del Cerro A, Alonso-Guervós M, Iglesias-Cabo T, et al. 2023. Generation of a novel three-dimensional scaffold-based model of the bovine endometrium. *Vet. Res. Commun.* 47(3):1721–33
164. Hill DS, Robinson NDP, Caley MP, Chen M, O'Toole EA, et al. 2015. A novel fully humanized 3D skin equivalent to model early melanoma invasion. *Mol. Cancer Ther.* 14(11):2665–73
165. Costello L, Fullard N, Roger M, Bradbury S, Dicolandrea T, et al. 2019. Engineering a multilayered skin equivalent: the importance of endogenous extracellular matrix maturation to provide robustness and reproducibility. In *Skin Tissue Engineering: Methods and Protocols*, ed. S Böttcher-Haberzeth, T Biedermann, pp. 107–22. New York: Springer
166. Totti S, Ng KW, Dale L, Lian G, Chen T, Velliou EG. 2019. A novel versatile animal-free 3D tool for rapid low-cost assessment of immunodiagnostic microneedles. *Sens. Actuators B* 296:126652
167. Arslan SY, Yu Y, Burdette JE, Pavone ME, Hope TJ, et al. 2015. Novel three dimensional human endocervix cultures respond to 28-day hormone treatment. *Endocrinology* 156(4):1602–9
168. De Bem THC, Tinning H, Vasconcelos EJR, Wang D, Forde N. 2021. Endometrium-on-a-chip reveals insulin- and glucose-induced alterations in the transcriptome and proteomic secretome. *Endocrinology* 162(6):bqab054
169. Xiao S, Coppeta JR, Rogers HB, Isenberg BC, Zhu J, et al. 2017. A microfluidic culture model of the human reproductive tract and 28-day menstrual cycle. *Nat. Commun.* 8:14584
170. Turco MY, Gardner L, Hughes J, Cindrova-Davies T, Gomez MJ, et al. 2017. Long-term, hormone-responsive organoid cultures of human endometrium in a chemically defined medium. *Nat. Cell Biol.* 19(5):568–77
171. Cindrova-Davies T, Zhao X, Elder K, Jones CJP, Moffett A, et al. 2021. Menstrual flow as a non-invasive source of endometrial organoids. *Commun. Biol.* 4:651
172. Kawasaki M, Dykstra GD, McConnel CS, Burbick CR, Ambrosini YM. 2023. Adult bovine-derived small and large intestinal organoids: *in vitro* development and maintenance. *J. Tissue Eng. Regen. Med.* 2023:e3095002
173. Yamauchi N, Yamada O, Takahashi T, Imai K, Sato T, et al. 2003. A three-dimensional cell culture model for bovine endometrium: regeneration of a multicellular spheroid using ascorbate. *Placenta* 24(2–3):258–69
174. Lawson EF, Ghosh A, Blanch V, Grupen CG, Aitken RJ, et al. 2023. Establishment and characterization of oviductal organoids from farm and companion animals. *Biol. Reprod.* 108(6):854–65
175. Turco MY, Gardner L, Kay RG, Hamilton RS, Prater M, et al. 2018. Trophoblast organoids as a model for maternal-fetal interactions during human placentation. *Nature* 564(7735):263–67
176. Davenport KM, O'Neil EV, Ortega MS, Patterson A, Kelleher AM, et al. 2024. Single-cell insights into development of the bovine placenta. *Biol. Reprod.* 110(1):169–84
177. Garcia-Flores V, Romero R, Tarca AL, Peyvandipour A, Xu Y, et al. 2024. Deciphering maternal-fetal cross-talk in the human placenta during parturition using single-cell RNA sequencing. *Sci. Transl. Med.* 16(729):eadh8335
178. Ming H, Zhang M, Rajput S, Logsdon D, Zhu L, et al. 2024. In vitro culture alters cell lineage composition and cellular metabolism of bovine blastocyst. *Biol. Reprod.* 111:11–27
179. Butt Z, Tinning H, O'Connell MJ, Fenn J, Alberio R, Forde N. 2023. Understanding conceptus-maternal interactions: What tools do we need to develop? *Reprod. Fertil. Dev.* 36(2):81–92
180. Adhikari B, Lee CN, Khadka VS, Deng Y, Fukumoto G, et al. 2022. RNA-sequencing based analysis of bovine endometrium during the maternal recognition of pregnancy. *BMC Genom.* 23:494
181. Kues WA, Sudheer S, Herrmann D, Carnwath JW, Havlicek V, et al. 2008. Genome-wide expression profiling reveals distinct clusters of transcriptional regulation during bovine preimplantation development in vivo. *PNAS* 105(50):19768–73
182. Mamo S, Mehta JP, McGettigan P, Fair T, Spencer TE, et al. 2011. RNA sequencing reveals novel gene clusters in bovine conceptuses associated with maternal recognition of pregnancy and implantation. *Biol. Reprod.* 85(6):1143–51

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44

183. Graf A, Krebs S, Zakhartchenko V, Schwalb B, Blum H, Wolf E. 2014. Fine mapping of genome activation in bovine embryos by RNA sequencing. *PNAS* 111(11):4139–44
184. Schalich KM, Koganti PP, Castillo JM, Reiff OM, Cheong SH, Selvaraj V. 2024. The uterine secretory cycle: recurring physiology of endometrial outputs that setup the uterine luminal microenvironment. *Physiol. Genom.* 56(1):74–97
185. Palma-Vera SE, Sharbati S, Einspanier R. 2015. Identification of miRNAs in bovine endometrium through RNAseq and prediction of regulated pathways. *Reprod. Domest. Anim.* 50(5):800–6
186. Yan C, Lv H, Peng Z, Yang D, Shen P, et al. 2021. Analysis of miRNA expression changes in bovine endometrial stromal cells treated with lipopolysaccharide. *Theriogenology* 167:85–93
187. Cuthbert JM, Russell SJ, Polejaeva IA, Meng Q, White KL, Benninghoff AD. 2021. Dynamics of small non-coding RNAs in bovine scNT embryos through the maternal-to-embryonic transition. *Biol. Reprod.* 105(4):918–33
188. Salilew-Wondim D, Tholen E, Held-Hoelker E, Shellander K, Blaschka C, et al. 2024. Endometrial DNA methylation signatures during the time of breeding in relation to the pregnancy outcome in post-partum dairy cows fed a control diet or supplemented with rumen-protected methionine. *Front. Genet.* 14:1267053
189. Chen Z, Hagen DE, Wang J, Elsik CG, Ji T, et al. 2016. Global assessment of imprinted gene expression in the bovine conceptus by next generation sequencing. *Epigenetics* 11(7):501–16
190. Rabaglino MB, O'Doherty A, Bojsen-Møller Secher J, Lonergan P, Hyttel P, et al. 2021. Application of multi-omics data integration and machine learning approaches to identify epigenetic and transcriptomic differences between *in vitro* and *in vivo* produced bovine embryos. *PLOS ONE* 16(5):e0252096
191. Choe C, Park J-W, Kim E-S, Lee S-G, Park S-Y, et al. 2010. Proteomic analysis of differentially expressed proteins in bovine endometrium with endometritis. *Korean J. Physiol. Pharmacol.* 14(4):205–12
192. Piras C, Guo Y, Soggiu A, Chanrot M, Greco V, et al. 2017. Changes in protein expression profiles in bovine endometrial epithelial cells exposed to *E. coli* LPS challenge. *Mol. BioSyst.* 13(2):392–405
193. Banliat C, Mahé C, Lavigne R, Com E, Pineau C, et al. 2022. The proteomic analysis of bovine embryos developed *in vivo* or *in vitro* reveals the contribution of the maternal environment to early embryo. *BMC Genom.* 23:839

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:13:44